

n -th Tropical Nevanlinna Theory Lean Blueprint

Abstract

Definitions, statements, and proofs follow the source paper in order. Each formalized item links to its verified Lean declaration; the compact formalization map beneath it identifies the implementation file.

2 Functions and multiplicities

Throughout, $n \in \mathbb{N}$ means $n \geq 1$. The max-plus operations are

$$a \oplus b = \max\{a, b\}, \quad a \otimes b = a + b, \quad a \oslash b = a - b.$$

Definition (Tropicalization). For a positive family $q(X, \varepsilon)$ and $n \geq 1$, write

$$q \xrightarrow[n]{\text{trop}} Q \iff \lim_{\varepsilon \rightarrow 0^+} \varepsilon^n \log q(X, \varepsilon) = Q(X) \quad \text{for every } X \in \mathbb{R}.$$

For finite nonempty supports I, J , degrees u_i, v_j , and real coefficients A_i, B_j , the dequantized rational factor is

$$R_\varepsilon(X) = \frac{\sum_{i \in I} \exp((u_i X + A_i)/\varepsilon)}{\sum_{j \in J} \exp((v_j X + B_j)/\varepsilon)}.$$

Its first tropicalization is

$$R(X) = \max_{i \in I} (u_i X + A_i) - \max_{j \in J} (v_j X + B_j).$$

An active product of n such factors, with the source parameter scaled so that the logarithm is multiplied by ε^n , is the paper's n -th tropical source.

Proposition (Formalized tropicalization rules). *Finite log-sum-exp tropicalizes to maximum. Rational numerator/denominator tropicalizes to the difference of their maxima, an active product of n factors tropicalizes to the sum of the factor limits, and tropicalizable positive families satisfy*

$$f + g \mapsto F \oplus G, \quad fg \mapsto F \otimes G, \quad f/g \mapsto F \oslash G.$$

Proof. If $M = \max_i a_i$, then

$$M \leq \varepsilon \log \sum_i e^{a_i/\varepsilon} \leq M + \varepsilon \log |I|,$$

so the squeeze theorem gives the finite log-sum-exp limit. Apply this once to the numerator and once to the denominator and subtract. Logarithms turn a finite product into a finite sum, so the n active factor limits add. For addition, factor out the larger positive summand; the remaining correction is between 0 and $\varepsilon^n \log 2$ and therefore vanishes. The product and quotient identities follow directly from $\log(fg) = \log f + \log g$ and $\log(f/g) = \log f - \log g$. $\square$

Definition (n -th tropical meromorphic functions). An n -th tropical meromorphic function is a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ admitting a locally finite, bi-infinite increasing mesh $(x_i)_{i \in \mathbb{Z}}$ and polynomial pieces p_i such that

$$f(x) = p_i(x) \quad (x \in [x_{i-1}, x_i]), \quad \deg p_i \leq n,$$

and at least one piece has degree n . The mesh is auxiliary; its redundant cut points are not singularities.

Formalization map. `PolynomialPresentation`, `IsNthTropicalMeromorphicFunction`, and `NthTropicalMeromorphicFunction` in `Function/PiecewisePolynomial.lean`. Intrinsic one-sided germs make all later multiplicities independent of the chosen presentation.

Definition (Normalized jets and jet matrix). For a polynomial $p(X) = \sum_{k=0}^n a_k X^k$, its normalized j -jet at x_0 is

$$J_j(p, x_0) = \frac{p^{(j)}(x_0)}{j!} = \sum_{k=j}^n \binom{k}{j} a_k x_0^{k-j}.$$

Writing $J(p, x_0) = (J_1, \dots, J_n)^T$ and $a = (a_1, \dots, a_n)^T$ gives $J(p, x_0) = \mathcal{C}_n(x_0)a$, where $\mathcal{C}_n(x_0)$ is upper triangular with diagonal entries 1 and hence is invertible. Multiplying its j -th row by the appropriate one-sided factor $\text{sgn}^{j+1}(x_0^\pm)$ gives the signed jet matrix $\mathcal{D}_n(x_0^\pm)$ used to recover prescribed multiplicities.

Definition (j -th roots, poles, and entire functions). For $1 \leq j \leq n$, define

$$\omega_f^{(j)}(x) = \frac{\text{sgn}^{j+1}(x^+)f^{(j)}(x^+) - \text{sgn}^{j+1}(x^-)f^{(j)}(x^-)}{j!}, \quad \text{sgn}(0^\pm) = \pm 1.$$

The point x is a j -th root if $\omega_f^{(j)}(x) > 0$, and a j -th pole if $\omega_f^{(j)}(x) < 0$; its multiplicity is $|\omega_f^{(j)}(x)|$. The function is *tropical entire* if it has no poles of any order, and *nowhere-vanishing entire* if it has neither roots nor poles.

Example (2.1 – Closure failures). Let $f(x) = \text{sgn}(x)x^2$ and $g(x) = x$. Both are second-order tropical entire, but $f \oplus g$ has a second-order pole of multiplicity 1 at 0, and $f(x+1)$ has a second-order pole of multiplicity 2 at -1 . Thus translation and max need not preserve higher-order entirety.

Proof. On the two sides of 0, the maximum $f \oplus g$ is respectively $-x^2$ and x near the origin. Substitution into the signed second-jet definition gives $\omega^{(2)}(0) = -1$. The translated function has its unique germ change at -1 ; its two second derivatives have signed jump $\omega^{(2)}(-1) = -2$. Both are therefore genuine second-order poles with the stated multiplicities. $\square$

Formalization map. `example_2_1_counterexamples` in `Counterexamples.lean`.

Proposition (2.2 – Endpoint mean inequality). If f is n -th tropical entire, then for every $r > 0$,

$$f(0) \leq \frac{f(r) + f(-r)}{2}.$$

If f is nowhere-vanishing entire, equality holds.

Proof. Taylor-expand the polynomial pieces from 0 to r and from 0 to $-r$. Adjacent polynomial values cancel, leaving the central polynomial jets and the weighted sums $\omega_f^{(j)}(x)(r - |x|)^j$. Every such term is nonnegative for an entire function. If there are no roots either, every jump vanishes, so the two endpoint expansions add to $2f(0)$. $\square$

Formalization map. `entire_midpoint_le_endpointMean` and `nowhereVanishingEntire_endpointMean_eq` in `Function/EntireMean.lean`.

Example (Counterexample to the converse of 2.2). Define $q(x) = 0$ for $|x| \leq 1$, $q(x) = -x + 1$ for $x > 1$, and $q(x) = -x - 1$ for $x < -1$. Then $q(r) + q(-r) = 2q(0)$ for every $r > 0$, but $x = 1$ is a genuine first-order pole. Thus the endpoint equality does not characterize tropical entirety.

Proof. The two tails are negatives of one another and the middle interval is zero, which gives the endpoint equality directly. At $x = 1$ the slope changes from 0 to -1 , hence $\omega_q^{(1)}(1) = -1 < 0$. $\square$

Formalization map. `proposition22_converse_is_false` in `Counterexamples.lean`.

Proposition (2.3 – Entire quotient decomposition). *Every n -th tropical meromorphic function f can be written $f = h \oslash g = h - g$, where g and h are tropical entire, their orders n_1, n_2 satisfy $\max\{n_1, n_2\} = n$, and they have no common j -th root for $1 \leq j \leq n$.*

Proof. At each genuine singularity prescribe the jump of g to be the positive part of $-\omega_f$ and put $h = f + g$. Then $\omega_g^{(j)} = \max\{-\omega_f^{(j)}, 0\}$ and $\omega_h^{(j)} = \max\{\omega_f^{(j)}, 0\}$. Polynomial-jet inversion and continuous spline assembly realize these prescribed jumps globally. Hence both functions are entire, have disjoint root data, and $h - g = f$. Exact ambient order follows because cancellation of every degree- n coefficient in both g and h would force the degree- n coefficient of f to vanish. $\square$

Formalization map. `exists_entire_quotient_decomposition` in `Function/Decomposition.lean`; jet inversion and assembly are encapsulated in `PolynomialJet.lean` and `SplineAssembly.lean`.

3 Poisson–Jensen and growth

Put

$$\tau_f^{(j)}(a) = \frac{f^{(j)}(a^+) - f^{(j)}(a^-)}{j!}.$$

Lemma (3.1 – Finite telescoping). *Let f be piecewise polynomial of degree at most n on $[-r, r]$, and $x \in (-r, r)$. If $x_1 < \dots < x_v$ are the genuine derivative-jump points in (x, r) and $x_{-\mu} < \dots < x_{-1}$ those in $(-r, x)$, then*

$$\begin{aligned} f(r) - f(x) &= \sum_{j=1}^n \left[\frac{f^{(j)}(x^+)}{j!} (r - x)^j + \sum_{i=1}^v \tau_f^{(j)}(x_i) (r - x_i)^j \right], \\ f(x) - f(-r) &= \sum_{j=1}^n \left[(-1)^{j+1} \frac{f^{(j)}(x^-)}{j!} (r + x)^j + (-1)^j \sum_{i=1}^{\mu} \tau_f^{(j)}(x_{-i}) (r + x_{-i})^j \right]. \end{aligned}$$

Proof. Taylor-expand each polynomial piece at its left endpoint. Add the increments along the mesh. Continuity cancels the zeroth-order terms, and exchanging the two finite sums collects the jump in each normalized derivative. The left formula follows identically after reversing orientation, producing the powers of -1 . $\square$

Formalization map. `endpoint_sub_eq_jet_sum`, `sub_leftEndpoint_eq_jet_sum`, and their genuine-singularity wrappers in `PoissonJensen/Telescoping.lean` and `PoissonJensen/IntrinsicTelescoping.lean`.

Definition (Poisson–Jensen kernels). For $r > 0$, $|x| < r$, and $k \geq 0$, define

$$B_k(r, x) = \frac{(r+x)^k + (r-x)^k}{2}, \quad D_k(r, x) = \frac{(r+x)^k - (r-x)^k}{2},$$

and for $|y| < r$ define

$$E(r, x, y) = r^2 - |x-y|r - xy = \begin{cases} (r-y)(r+x), & x \leq y, \\ (r+y)(r-x), & y \leq x. \end{cases}$$

Thus $B_k \pm D_k = (r \pm x)^k$, $D_k(r, 0) = 0$, and $E(r, 0, y) = r(r - |y|)$.

Definition (Nine-region partition). For fixed $r > 0$ and $|x| < r$, let

$$\begin{aligned} F_1 &= \{y : -r < y < \min(0, x)\}, & F_2 &= \{y : \max(0, x) < y < r\}, \\ F_3 &= \{y : y = x = 0\}, & F_4 &= \{y : 0 < y < x\}, \\ F_5 &= \{y : x < y < 0\}, & F_6 &= \{y : x < y = 0\}, \\ F_7 &= \{y : y = 0 < x\}, & F_8 &= \{y : 0 < y = x\}, \\ F_9 &= \{y : y = x < 0\}. \end{aligned}$$

They are disjoint and cover $(-r, r)$; some are empty for a fixed x .

Definition (Auxiliary jumps). For $1 \leq j \leq n$ define

$$\begin{aligned} \Omega_f^{(j)}(x, y) &= \frac{\operatorname{sgn}^{j+1}((y-x)^+)f^{(j)}(y^+) - \operatorname{sgn}^{j+1}((y-x)^-)f^{(j)}(y^-)}{j!}, \\ \Gamma_f^{(j)}(x, y) &= \frac{\operatorname{sgn}^j((y-x)^+)f^{(j)}(y^+) - \operatorname{sgn}^j((y-x)^-)f^{(j)}(y^-)}{j!}. \end{aligned}$$

In particular $\Omega_f^{(j)}(0, y) = \omega_f^{(j)}(y)$.

Lemma (3.2 – Nine-region reduction). For $y \in F_\ell$, write $L_j(y) = f^{(j)}(y^-)/j!$. Then

$$\Omega_f^{(j)}(x, y) = A_{\ell,j}\omega_f^{(j)}(y) + A'_{\ell,j}L_j(y), \quad \Gamma_f^{(j)}(x, y) = C_{\ell,j}\omega_f^{(j)}(y) + C'_{\ell,j}L_j(y),$$

where

ℓ	1	2	3	4	5	6	7	8	9
$A_{\ell,j}$	1	1	1	$(-1)^{j+1}$	$(-1)^{j+1}$	1	$(-1)^{j+1}$	1	$(-1)^{j+1}$
$A'_{\ell,j}$	0	0	0	0	0	$(-1)^{j+1} - 1$	$1 - (-1)^{j+1}$	$1 - (-1)^{j+1}$	$1 - (-1)^{j+1}$
$C_{\ell,j}$	-1	1	1	$(-1)^j$	$(-1)^{j+1}$	1	$(-1)^j$	1	$(-1)^{j+1}$
$C'_{\ell,j}$	0	0	$2(-1)^{j+1}$	0	0	$(-1)^{j+1} - 1$	$(-1)^{j+1} - 1$	$1 - (-1)^j$	$1 - (-1)^j$

Proof. Put $u = f^{(j)}(y^+)/j!$, $v = f^{(j)}(y^-)/j!$, and $\varepsilon = (-1)^{j+1}$. The multiplicity definition gives

$$\omega_f^{(j)}(y) = \begin{cases} u - v, & y > 0, \\ \varepsilon(u - v), & y < 0, \\ u - \varepsilon v, & y = 0. \end{cases}$$

The sign of $y - x$ similarly gives

$$\Omega_f^{(j)}(x, y) = \begin{cases} u - v, & y > x, \\ \varepsilon(u - v), & y < x, \\ u - \varepsilon v, & y = x, \end{cases}$$

while Γ is obtained from the same three rows after replacing the right-side exponent $j + 1$ by j . Each region fixes both the sign of y and the comparison of y with x . Solving the first display for $u - v$ and substituting into the second gives the four coefficients in the table. The cases F_6, F_7, F_8, F_9 retain a multiple of v because either $y = 0$ or $y = x$; in all other cases that correction is zero. Direct substitution for the nine disjoint regions yields every table entry. $\square$

Formalization map. `auxiliaryJump_eq_byRegion` in `PoissonJensen/AuxiliaryJumps.lean`. The definitions B_k, D_k, E and the nine-region partition remain proof-local in `Kernels.lean` and `RegionPartition.lean`.

Theorem (3.3 – Poisson–Jensen formula). *Let $r > 0$, $|x| < r$, and let S be the finite set of genuine singularities of f in $(-r, r)$. If $y \in F_\ell$, put $A_j(x, y) = A_{\ell,j}$ and $C_j(x, y) = C_{\ell,j}$ from Lemma 3.2. Let f_a- denote the polynomial germ immediately to the left of a , extended as a polynomial to all real arguments, and let $s_-(x) = \text{sgn}(x)$ for $x \neq 0$ and $s_-(0) = -1$. Then*

$$\begin{aligned} f(x) = & \frac{f(r) + f(-r)}{2} + \frac{D_n(r, x)}{2B_n(r, x)}(f(r) - f(-r)) \\ & - \sum_{j=1}^n \sum_{y \in S} A_j(x, y) \omega_f^{(j)}(y) E(r, x, y)^j \frac{B_{n-j}(r, x)}{2B_n(r, x)} \\ & - \sum_{j=1}^n \sum_{y \in S} C_j(x, y) \omega_f^{(j)}(y) E(r, x, y)^j \frac{D_{n-j}(r, x)}{2B_n(r, x)} \\ & - \frac{(r+x)^n}{2B_n(r, x)}(f_{x^-}(r) - f(x)) - \frac{(r-x)^n}{2B_n(r, x)}(f_{x^-}(-r) - f(x)) \\ & - s_-(x) \frac{(r-|x|)^n}{2B_n(r, x)}(f_0^-(r) + f_0^-(-r) - 2f(0)). \end{aligned}$$

At $x = 0$ all D -terms and germ corrections cancel, giving

$$f(0) = \frac{f(r) + f(-r)}{2} - \frac{1}{2} \sum_{j=1}^n \sum_{\substack{y: j\text{-root} \\ |y| < r}} |\omega_f^{(j)}(y)| (r - |y|)^j + \frac{1}{2} \sum_{j=1}^n \sum_{\substack{z: j\text{-pole} \\ |z| < r}} |\omega_f^{(j)}(z)| (r - |z|)^j.$$

Proof. Multiply the two identities of Lemma 3.1 by $(r+x)^n$ and $(r-x)^n$ and subtract. Factor every distance product through $E(r, x, y)$, split symmetric and antisymmetric powers into B_k and D_k , and use Lemma 3.2 to replace auxiliary jumps by intrinsic multiplicities. Taylor’s formula recombines the remaining one-sided jets into the two germ corrections. Division by $2B_n(r, x) > 0$ yields the general formula. Substitution $x = 0$ leaves the signed root/pole sum above. $\square$

Formalization map. `poissonJensen` and `jensenFormula` in `PoissonJensen/Main.lean`. The public theorem constructs the needed finite interval presentation from the global function.

Example (3.4 – Multiplicity table). On $(-3, 3)$ let

$$f(x) = \begin{cases} -2x^3 - 17x^2 - 46x - 40, & -3 < x \leq -2, \\ 2x^2 + 6x + 4, & -2 < x \leq -1, \\ x^2 + 2x + 1, & -1 < x \leq 1, \\ -x + 5, & 1 < x \leq 2, \\ 3x^3 - 15x^2 + 18x + 3, & 2 < x < 3. \end{cases}$$

Its nonzero signed multiplicities are

x	-2	-1	0	1	2
$\omega_f^{(1)}$		-2		-5	-5
$\omega_f^{(2)}$	-7	1	2	-1	3
$\omega_f^{(3)}$	2				3

and for $2 < r < 3$ the third-order Jensen formula evaluates to $f(0) = 1$.

Proof. Differentiate the two polynomial germs at each of $-2, -1, 0, 1, 2$, divide the j -th derivative jump by $j!$, and insert the one-sided sign factors. This gives exactly the displayed table; blank entries are zero. Separating its positive and negative entries into roots and poles and substituting their weights $(r - |x|)^j/2$ into the Jensen formula cancels every cubic, quadratic, and linear term, leaving $1 = f(0)$. $\square$

Formalization map. `example34_complete_multiplicity_table` in `Counterexamples.lean`.

Definition (Nevanlinna quantities). For $r > 0$ define

$$m(r, f) = \frac{\max\{f(r), 0\} + \max\{f(-r), 0\}}{2},$$

$$N^{(j)}(r, f) = \frac{1}{2} \sum_{z: j\text{-pole}, |z| < r} |\omega_f^{(j)}(z)| (r - |z|)^j,$$

$$T(r, f) = m(r, f) + \sum_{j=1}^n N^{(j)}(r, f).$$

For a region $A \subseteq \mathbb{R}$, $N_A^{(j)}(r, f)$ is the same finite sum restricted to $z \in A$. If $n^{(j)}(t, f)$ counts j -th poles in $(-t, t)$, then

$$N^{(j)}(r, f) = \frac{1}{2} \int_{[0, r]^j} n^{(j)}\left(\min_s t_s, f\right) dt_1 \cdots dt_j.$$

Finally

$$\rho(f) = \limsup_{r \rightarrow \infty} \frac{\log T(r, f)}{\log r}, \quad \rho_2(f) = \limsup_{r \rightarrow \infty} \frac{\log \log T(r, f)}{\log r}.$$

Sum-integral identity. Expand the counting function into the finite sum of indicators of boxes $[|z|, r]^j$. Tonelli reduces the integral to a finite sum, and the box attached to z has volume $(r - |z|)^j$. $\square$

Formalization map. Definitions `proximity`, `maxPlusCounting`, `integratedCounting`, `partialIntegratedCounting`, and `characteristic`; the bridge and integral identity are `countingBoxIntegrand_eq_maxPlusCounting_minimum`, `integratedCounting_eq_iteratedIntegral`, and `integratedCounting_eq_paperIteratedIntegral` in `Nevanlinna/Counting.lean`. Growth invariants are in `Nevanlinna/Growth.lean`.

Theorem (3.5 – Jensen identity). For every $r > 0$,

$$T(r, f) = T(r, -f) + f(0).$$

Proof. Split the signed multiplicity sum in the Jensen formula into positive roots and negative poles. Replacing f by $-f$ exchanges the two sets, while $\max(a, 0) - \max(-a, 0) = a$ converts the endpoint term into $(f(r) + f(-r))/2$. Rearrangement gives the identity. $\square$

Formalization map. [characteristic_eq_neg_add_at_zero](#) in [Nevanlinna/Jensen.lean](#).

Definition (3.6–3.7 – Tropical hyper-exponentials). For $\alpha > 1$, the first-order tropical hyper-exponential is

$$e_\alpha(x) = \alpha^{\lfloor x \rfloor} \left(x - \lfloor x \rfloor + \frac{1}{\alpha - 1} \right).$$

For $n \geq 1$ define

$$\begin{aligned} e_{n,\alpha}(x) = & \operatorname{sgn}^{n+1}(x^+) \alpha^{\lfloor x \rfloor} (x^n - \lfloor x \rfloor^n) \\ & + \sum_{i=-\infty}^{\lfloor x \rfloor - 1} \operatorname{sgn}^{n+1}(i^+) \alpha^i ((i+1)^n - i^n). \end{aligned}$$

The negative infinite tail is interpreted as its absolutely convergent geometric-polynomial series. For $n = 1$ this reduces to e_α .

Proposition (3.8 – Hyper-exponential growth). For $\alpha > 1$, the generalized tropical hyper-exponential $e_{n,\alpha}$ is a nonnegative, nondecreasing, n -th tropical entire function and $\rho_2(e_{n,\alpha}) = 1$.

Proof. Reindex the negative tail by $k = -i$; it is bounded by a finite linear combination of series $\sum_{k \geq 0} k^q \alpha^{-k}$ and therefore converges absolutely. At an integer m , the left polynomial contributes precisely the $i = m - 1$ term missing from the right tail, so the two one-sided limits agree. On a nonintegral interval the derivative is $n\alpha^{\lfloor x \rfloor} |x|^{n-1} \geq 0$, proving monotonicity; the limit at $-\infty$ and monotonicity give nonnegativity.

At an integer, direct normalized-jet computation gives nonnegative $\omega_{e_{n,\alpha}}^{(j)}$ for every $1 \leq j \leq n$, while away from the integers every multiplicity is zero. Hence the function is tropical entire. For large positive r , its endpoint value is bounded above by $C(1+r)^n \alpha^r$ and below on a fixed terminal part of each unit interval by $c\alpha^{r-1}$. Jensen and nonnegativity compare these bounds with the characteristic. Taking two logarithms and dividing by $\log r$ squeezes the limsup between 1 and 1, so $\rho_2(e_{n,\alpha}) = 1$. $\square$

Formalization map. [tropicalHyperExponential](#) and [proposition_3_8](#) in [Nevanlinna/Examples.lean](#).

4 Logarithmic differences

Example (4.1 – Failure of monotonicity and convexity). For every $m \in \mathbb{N} \cup \{0\}$, define

$$f(x) = \begin{cases} -(x-m)(x-m-1), & m \leq x \leq m+1, \\ (x+m)(x+m+1), & -m-1 \leq x \leq -m. \end{cases}$$

Then

$$\begin{aligned} m(r, f) &= -\frac{1}{2}(r - \lfloor r \rfloor)(r - \lfloor r \rfloor - 1), \\ \sum_{j=1}^2 N^{(j)}(r, f) &= N^{(1)}(r, f) = \frac{1}{2} \sum_{i=1}^{\lfloor r \rfloor} 2(r-i), \\ T(r, f) &= -\frac{1}{2}r^2 + \left(2\lfloor r \rfloor + \frac{1}{2}\right)r - \lfloor r \rfloor(\lfloor r \rfloor + 1). \end{aligned}$$

In particular, $T(r, f)$ is neither nondecreasing nor convex.

Proof. On each unit interval the endpoint values give the displayed proximity term. The first derivatives jump down by 2 at every nonzero integer of the appropriate sign, while the normalized second jets agree; hence only the first-order counting term remains and summing its weights gives the second formula. Their sum is the displayed expression for T . On $0 \leq r \leq 1$ it reduces to $(-r^2 + r)/2$: it decreases after $r = 1/2$ and has second derivative -1 . Thus it is neither nondecreasing nor convex. $\square$

Formalization map. `example41_characteristic_not_monotone` and `example41_characteristic_not_convex` in `Counterexamples.lean`.

Definition (4.2 – Well-defined polynomials). A polynomial is *well-defined* if, after subtracting its value at zero, it contains only odd powers when its degree is odd or only even powers when its degree is even, and all nonzero coefficients have one common weak sign. An n -th tropical meromorphic function is well-defined when each polynomial piece is well-defined.

Formalization map. `IsWellDefinedPolynomial` and `IsWellDefinedNthTropicalMeromorphicFunction` in `LogDerivative/WellDefined.lean`.

Lemma (4.3 – Radial derivative signs). Let p be a well-defined polynomial of degree n , $\delta \in \{-1, 1\}$, and $r > 0$. For all $1 \leq \ell \leq n$, the derivatives $d^\ell p(\delta r)/dr^\ell$ have one common weak sign.

Proof. If $n = 2m$, then

$$\frac{d^\ell}{dr^\ell} p(\delta r) = \sum_{2j \geq \ell} a_{2j} \frac{(2j)!}{(2j - \ell)!} r^{2j - \ell}.$$

Every weight is nonnegative, so the sum has the common coefficient sign. If $n = 2m - 1$, the same calculation gives

$$\frac{d^\ell}{dr^\ell} p(\delta r) = \delta \sum_{2j-1 \geq \ell} a_{2j-1} \frac{(2j-1)!}{(2j-1-\ell)!} r^{2j-1-\ell},$$

whose sign is the coefficient sign multiplied by the fixed δ . $\square$

Formalization map. `wellDefinedPolynomial_iteratedDeriv_commonSign` in `LogDerivative/WellDefined.lean`.

Definition (Well-defined n -th tropical meromorphic function). Such a function is well-defined when every polynomial germ in one, hence any, intrinsic piecewise-polynomial realization is well-defined in Definition 4.2.

Lemma (4.4 – Characteristic monotonicity). For a well-defined n -th tropical meromorphic function f , every left radial derivative $T^{(\ell)}(r^-, f)$, $0 \leq \ell \leq n$, is nonnegative and nondecreasing. In particular $T(\cdot, f)$ is nonnegative, nondecreasing, and convex on $(0, \infty)$.

Proof. Put $h_\delta(s) = f(\delta s)$. At an event radius $a > 0$, write $u_\delta^\pm = h_\delta^{(\ell)}(a^\pm)$ and let $\chi_\delta^\pm$ indicate that h_δ is positive on the corresponding side. The multiplicity definition gives

$$u_\delta^+ - u_\delta^- = \ell! \omega_f^{(\ell)}(\delta a).$$

The combined jump of the proximity and pole-counting terms in the ℓ -th derivative is

$$J_{\delta, \ell}(a) = \frac{1}{2} (\chi_\delta^+ u_\delta^+ - \chi_\delta^- u_\delta^- + \max\{u_\delta^- - u_\delta^+, 0\}).$$

If $h_\delta(a) > 0$, this is $\max\{u_\delta^+ - u_\delta^-, 0\}/2$; if $h_\delta(a) < 0$, it is $\max\{u_\delta^- - u_\delta^+, 0\}/2$. If $h_\delta(a) = 0$, Lemma 4.3 gives $\chi_\delta^+ u_\delta^+ \geq 0$ and $-\chi_\delta^- u_\delta^- \geq 0$. Thus every derivative jump is nonnegative.

Near the origin, Taylor expansion gives

$$T(r, f) = T(0^+, f) + \sum_{j=1}^n c_j r^j, \quad c_j = \frac{\chi_+ a_j + \chi_- b_j + \max\{-(a_j + b_j), 0\}}{2j!},$$

where $a_j = d^j f(r)/dr^j|_{0^+}$ and $b_j = d^j f(-r)/dr^j|_{0^+}$. If $f(0)$ is positive or negative this reduces respectively to the positive or negative part of $a_j + b_j$; if $f(0) = 0$, Lemma 4.3 controls the two indicated terms separately. Hence $c_j \geq 0$.

Between locally finite event radii, T is a polynomial of degree at most n . Its n -th derivative is therefore constant there and has only nonnegative jumps, so it is nonnegative and nondecreasing. Descending in ℓ , integrate between events and use the nonnegative jumps and the initial coefficients. This proves the assertion for every ℓ ; the cases $\ell = 0, 1$ give monotonicity and convexity. $\square$

Formalization map. `characteristicLemma44` in `LogDerivative/CharacteristicMonotonicity.lean`.

Lemma (4.5 – Pointwise shift estimate). *If f is well-defined, $\alpha > 1$, $c \neq 0$, $\delta \in \{-1, 1\}$, and*

$$r > \max\left\{2|c|, \frac{3-\alpha}{\alpha-1}|c|\right\},$$

then

$$|f(\delta r + c) - f(\delta r)| \leq \frac{32|c|}{(\alpha-1)(r+|c|)} \left(T(\alpha(r+|c|), f) + \frac{|f(0)|}{2} \right).$$

Proof. Set $\rho = (\alpha+1)(r+|c|)/2$. Apply Lemma 3.1 on the interval from δr to $\delta r + c$ and again out to $\delta\rho$. Common-sign jets permit the short increments to be bounded by the long increments. Split each singularity contribution into root and pole parts, dominate the partial counts by the full counts at ρ , and use the Jensen identity to replace the characteristic of $-f$. Monotonicity from Lemma 4.4 and the definition of ρ give the displayed constant. $\square$

Formalization map. `pointwiseShiftEstimate_of_wellDefined` in `LogDerivative/ShiftEstimate.lean`.

Lemma (4.6 – Growth shift). *Let $T : [0, \infty) \rightarrow [0, \infty)$ be continuous and nondecreasing with hyper-order $\rho_2 < 1$. If $c > 0$ and $0 < \sigma < 1 - \rho_2$, then outside a set of finite logarithmic measure,*

$$T(r+c) = T(r) + o\left(\frac{T(r)}{r^\sigma}\right).$$

Proof. Choose $\tilde{\sigma}$ with $\sigma < \tilde{\sigma} < 1 - \rho_2$. If T eventually vanishes there is nothing to prove; otherwise take $R > \max\{1, c\}$ with $T(R) > 0$. For $\eta > 0$ set

$$F_\eta = \left\{ r \geq R : \frac{T(r+c) - T(r)}{T(r)} r^{\tilde{\sigma}} \geq \eta \right\}.$$

We prove that F_η has finite logarithmic measure. Suppose instead that $\int_{F_\eta} dt/t = \infty$. Continuity makes F_η closed and it must be unbounded. Recursively choose

$$r_0 = \min F_\eta, \quad r_{m+1} = \min(F_\eta \cap [r_m + c, \infty)).$$

Then

$$F_\eta \subseteq \bigcup_{m \geq 0} [r_m, r_m + c], \quad T(r_{m+1}) \geq \left(1 + \frac{\eta}{r_m^{\tilde{\sigma}}}\right) T(r_m). \quad (4.6.1)$$

For every $\varepsilon > 0$, infinitely many m satisfy $r_m < m^{1+\varepsilon}$. Otherwise, from the covering and $\log(1+x) \leq x$,

$$\int_{F_\eta} \frac{dt}{t} \leq C + \sum_{m \gg 1} \log \frac{r_m + c}{r_m} \leq C + c \sum_{m \gg 1} m^{-1-\varepsilon} < \infty,$$

a contradiction. Choose ε so that $1/(1+\varepsilon) > \tilde{\sigma}$ and pass to such a subsequence. Iterating (4.6.1), using $r_k \leq r_m$, and then $\log(1+x) \geq x/2$ for small $x > 0$, gives

$$\log T(r_m) \geq C_0 + \frac{\eta m}{2r_m^{\tilde{\sigma}}}.$$

The right side tends to infinity, and a second logarithm yields

$$\frac{\log \log T(r_m)}{\log r_m} \geq \frac{\log m}{\log r_m} - \tilde{\sigma} + o(1) \geq \frac{1}{1+\varepsilon} - \tilde{\sigma} + o(1).$$

Thus $\rho_2 \geq 1/(1+\varepsilon) - \tilde{\sigma}$; letting $\varepsilon \downarrow 0$ contradicts $\tilde{\sigma} < 1 - \rho_2$. Hence F_η has finite logarithmic measure.

Take $E = F_1$. For $r \notin E$,

$$0 \leq T(r+c) - T(r) < \frac{T(r)}{r^{\tilde{\sigma}}} = \frac{T(r)}{r^\sigma} r^{-(\tilde{\sigma}-\sigma)},$$

and the final factor tends to zero. This is the claimed estimate. $\square$

Formalization map. `growthShiftLemma` in `LogDerivative/GrowthShiftProof.lean`.

Theorem (4.7 – Pointwise logarithmic derivative). *If f is well-defined, $c \neq 0$, $\rho_2(f) < 1$, and $0 < \sigma < 1 - \rho_2(f)$, then there is a set E of finite logarithmic measure such that, for both $\delta = \pm 1$,*

$$|f(\delta r + c) - f(\delta r)| = o\left(\frac{T(r, f)}{r^\sigma}\right) \quad (r \rightarrow \infty, r \notin E).$$

Proof. We first prove the power-step estimate used below. If T is nondecreasing, has hyper-order ρ_2 , and $0 < q < 1 - \rho_2$, then outside a set E_q of finite logarithmic measure,

$$T(r + r^q) \leq 2T(r). \tag{4.7.1}$$

Choose $\rho_2 < \beta < 1 - q$. Eventually $|\log T(x)| \leq x^\beta$ after discarding the identically-zero case. Put $U = \log T$ and let $F = \{x : U(x + x^q) - U(x) \geq \log 2\}$. On the exponential annulus $[a_m, b_m] = [Re^m, Re^{m+1}]$, set $H_m = b_m^q$. Monotonicity and the sliding-slope identity give

$$\begin{aligned} \int_{F \cap [a_m, b_m]} \frac{dx}{x} &\leq \frac{H_m}{(\log 2)a_m} (U(b_m + H_m) - U(a_m)) \\ &\leq C a_m^{q+\beta-1}. \end{aligned}$$

The geometric series converges because $q + \beta - 1 < 0$. Hence F has finite logarithmic measure, and exponentiation outside F proves (4.7.1).

Apply this to $T(r, f)$ with $q = (\sigma + 1 - \rho_2)/2$, so $\sigma < q < 1 - \rho_2$. For large $r \notin E_q$, let $s = r + |c|$ and $A_r = (r + r^q)/s$. Then

$$(A_r - 1)s = r^q - |c|, \quad A_r s = r + r^q.$$

Eventually $A_r > 1$ and the radius hypothesis of Lemma 4.5 holds. Lemma 4.5 and (4.7.1) therefore imply, simultaneously for $\delta = \pm 1$,

$$|f(\delta r + c) - f(\delta r)| \leq \frac{32|c|}{r^q - |c|} \left(T(r + r^q, f) + \frac{|f(0)|}{2} \right) \leq C \frac{T(r, f)}{r^q}.$$

Here, if $T(R_0, f) > 0$, monotonicity absorbs $|f(0)|$ into $T(r, f)$. Since $q > \sigma$, the last quantity is $r^{\sigma-q}T(r, f)/r^\sigma = o(T(r, f)/r^\sigma)$.

If $T(r, f) \equiv 0$, apply Lemma 4.5 with any sufficiently large constant A in place of α and let $A \rightarrow \infty$; this forces every displayed shift difference to be zero. Thus the same conclusion holds in the degenerate case, and one common exceptional set works for both signs. $\square$

Corollary (4.8 – Logarithmic derivative lemma). *Under the same hypotheses,*

$$m(r, f(x+c) \oslash f(x)) = o\left(\frac{T(r, f)}{r^\sigma}\right)$$

outside a set of finite logarithmic measure.

Proof. At the two radial endpoints, $u^+ \leq |u|$. Average the two estimates from Theorem 4.7. $\square$

Formalization map. `pointwiseLogarithmicDerivative` and `logarithmicDerivativeProximity` in `LogDerivative/Main.lean`.

5 Holomorphic curves

Definition (Tropical holomorphic curves). An n -th tropical holomorphic curve is a projective class $f = [f_0 : \dots : f_m]$ of tropical entire coordinates whose maximum order is n . A representation is reduced when the coordinates have no common j -th root. With $F(x) = \max_i f_i(x)$, define

$$T_f(r) = \frac{F(r) + F(-r)}{2} - F(0).$$

Formalization map. `TropicalHolomorphicCurveRepresentation`, `NthTropicalHolomorphicCurve`, and `cartanCharacteristic` in `Curves/Basic.lean` and `Curves/Characteristic.lean`.

Proposition (5.2 – Representation invariance). *The Cartan characteristic is independent of the reduced representation.*

Proof. Two reduced representatives differ by a common function λ . A root or pole of λ would create a common root after adding or subtracting it from every coordinate, contradicting reducedness. Thus λ is nowhere-vanishing entire. Proposition 2.2 gives $(\lambda(r) + \lambda(-r))/2 = \lambda(0)$, so the common additive change cancels from the defining expression for T_f . $\square$

Proposition (5.3 – Projective-pair characteristic). *If $g = g_1 - g_0$ and $[g_0 : g_1]$ is reduced, then*

$$T_{[g_0 : g_1]}(r) = T(r, g) - g^+(0).$$

Proof. Since $\max(g_0, g_1) = g_0 + (g_1 - g_0)^+$, substitute this identity into the Cartan characteristic. Jensen applied to g_0 replaces its endpoint mean minus $g_0(0)$ by the difference of its root and pole counts. Reducedness identifies those counts with the corresponding pole counts of $g_1 - g_0$; the remaining endpoint term is $m(r, g)$, yielding the formula. $\square$

Proposition (5.4 – Coordinate quotient bound). *For every coordinate quotient of a reduced curve,*

$$T(r, f_i - f_\ell) \leq T_f(r) + O(1).$$

Proof. Construct the common-root part u of f_i and f_ℓ by taking the minimum of each positive multiplicity and fixing $u(0) = f_\ell(0)$. Then $h_i = f_i - u$ and $h_\ell = f_\ell - u$ form a reduced pair. Proposition 5.3 and $\max(h_i, h_\ell) = \max(f_i, f_\ell) - u$ give

$$T(r, f_i - f_\ell) \leq \frac{1}{2} \sum_{\delta=\pm 1} \left(\max_k f_k(\delta r) - u(\delta r) \right) + O(1).$$

Because u is entire, Proposition 2.2 bounds its negative endpoint mean by a constant. The right side is therefore at most $T_f(r) + O(1)$. $\square$

Formalization map. `cartanCharacteristic_eq_of_reducedCurveRepresentations`, `cartanCharacteristic_projectivePair_eq_characteristic_sub`, and `characteristic_coordinateQuotient_le_upToConstant` in `Curves/Characteristic.lean`.

Definition (Homogeneous and Fermat tropical polynomials). A homogeneous tropical polynomial of degree d is a finite maximum of monomials

$$P(x_0, \dots, x_m) = \max_{|I|=d} \left(\alpha_I + \sum_{k=0}^m I_k x_k \right), \quad |I| = \sum_k I_k = d.$$

Composition is $(P \circ f)(x) = \max_{|I|=d} (\alpha_I + \sum_k I_k f_k(x))$. A tropical Fermat polynomial retains only the pure terms $\max_i (\alpha_i + dx_i)$.

Example (5.5 – Counting dominates curve growth). Let $P(x_0, x_1) = x_0 \oplus x_1$, $f_0(x) = \text{sgn}(x)x^2$, and

$$f_1(x) = \begin{cases} -\frac{1}{2}, & x \leq 0, \\ (4k+2)x - 2k(2k+2) - \frac{1}{2}, & 2k \leq x < 2k+2, \quad k \geq 0. \end{cases}$$

Then $T_f(r) = \frac{1}{2}r^2(1 + o(1))$, whereas

$$\sum_{j=1}^2 N^{(j)}(r, P \circ f) \geq \frac{1}{2} \sum_{i=1}^{\lfloor r/2 \rfloor} (r - 2i)^2 = \frac{r^3}{12}(1 + o(1)).$$

Consequently $T_f = o(\sum_j N^{(j)}(r, P \circ f))$.

Proof. On $[2k, 2k+2)$, solving $x^2 = f_1(x)$ gives $x = 2k+1 \pm \sqrt{2}/2$. At the lower crossing the maximum switches from the affine branch to the quadratic branch and its normalized second-jet jump is -1 , so this is a genuine second-order pole of multiplicity 1. Summing their integrated weights gives the displayed lower bound. The maximum of the two coordinates has quadratic endpoint growth $r^2 + o(r^2)$ and vanishes at the central normalization up to a constant, giving $T_f(r) = r^2/2 + o(r^2)$. Cubic growth dominates quadratic growth. $\square$

Formalization map. `example55_characteristic_isLittleO_integratedCounting` in `Counterexamples.lean`.

Theorem (5.6 – Tropical polynomial second main theorem). *Let P be homogeneous of degree $d > 0$ and contain every pure power with a finite coefficient. For a nonconstant reduced curve,*

$$T_f(r) = \frac{1}{d} \left(\sum_{j=1}^n N^{(j)}(r, -P \circ f) - \sum_{j=1}^n N^{(j)}(r, P \circ f) \right) + O(1).$$

Proof. The pure terms and the maximum coefficient give uniform bounds

$$\gamma + d \max_i f_i(x) \leq P \circ f(x) \leq \beta + d \max_i f_i(x).$$

Apply Jensen to $P \circ f$:

$$\sum_j N^{(j)}(r, -P \circ f) - \sum_j N^{(j)}(r, P \circ f) = \frac{P \circ f(r) + P \circ f(-r)}{2} - P \circ f(0).$$

Average the two pointwise bounds. The right side equals $dT_f(r) + O(1)$; division by d proves the formula. $\square$

Corollary (5.7 – First-order Fermat case). *For a nonconstant reduced first-order curve and a tropical Fermat polynomial of degree $d > 0$,*

$$T_f(r) = \frac{1}{d} N^{(1)}(r, -P \circ f) + O(1).$$

Proof. Every first-order entire coordinate is convex piecewise linear; multiplication by $d > 0$, addition of constants, and a finite maximum preserve convexity. Thus $P \circ f$ is tropical entire and its pole count vanishes. Apply Theorem 5.6. $\square$

Formalization map. `secondMain_homogeneousTropicalPolynomial` and `secondMain_firstOrder_tropicalFermat` in `Curves/SecondMain.lean`. Composition closure is constructed, not assumed.

Theorem (5.8 – Ordinary Fermat estimate). *Let f be a nonconstant reduced first-order curve and $\mathcal{P}(x) = \sum_{i=0}^m \alpha_i x_i^n$ with $n \geq 1$ and $\alpha_i > 0$. If $r/T_f(r) \rightarrow 0$, then*

$$\theta T_f(r)^n \leq \sum_{j=1}^n N^{(j)}(r, -\mathcal{P} \circ f) + o(T_f(r)^n) \leq \Theta T_f(r)^n,$$

where $\theta = \min_i \alpha_i$ and $\Theta = 2^{n-1} \sum_i \alpha_i$.

Proof. On a linear germ $g(x) = kx + b$,

$$\frac{(g^n)^{(j)}}{j!} = \binom{n}{j} g^{n-j} k^j,$$

so every singularity of f_i^n lies at a singularity of f_i or at the origin. A first-order entire coordinate is convex and has one of two tail types: its two slopes eventually point outward, or all slopes have one weak sign. In the first case all sufficiently remote jumps of f_i^n are roots, and the compact remainder is $O(r^n)$. In the second, at most one sign crossing occurs; comparison with the affine tail and first-order Jensen gives a total remote contribution $O(r^n)$. Since $r = o(T_f)$,

$$\sum_{j=1}^n N^{(j)}(r, f_i^n) = o(T_f(r)^n).$$

Subadditivity of pole counting and positive homogeneity give the same estimate for $\mathcal{P} \circ f$.

Jensen now yields

$$\sum_{j=1}^n N^{(j)}(r, -\mathcal{P} \circ f) = \frac{1}{2} \sum_{\delta=\pm 1} \mathcal{P} \circ f(\delta r) + o(T_f(r)^n).$$

Put $F = \max_i f_i$. The affine tails just analyzed show that any negative endpoint is $o(T_f)$, while $T_f = (F(r) + F(-r))/2 + O(1)$. Convexity of the n -th power and $a^n + b^n \leq (a + b)^n$ for nonnegative a, b give

$$\theta T_f(r)^n + o(T_f^n) \leq \frac{1}{2} \sum_{\delta=\pm 1} \sum_i \alpha_i f_i(\delta r)^n \leq 2^{n-1} \left(\sum_i \alpha_i \right) T_f(r)^n + o(T_f^n).$$

Combining the last two displays proves both inequalities. $\square$

Formalization map. [secondMain_ordinaryFermat](#) in [Curves/Fermat.lean](#).

Example (5.9 – Necessity of the growth hypothesis). For $f = [0 : 2x]$ and $\mathcal{P}(x_0, x_1) = x_0^2 + 2x_1^2$,

$$T_f(r) = r, \quad \mathcal{P} \circ f = 8x^2, \quad \sum_{j=1}^2 N^{(j)}(r, -\mathcal{P} \circ f) = 8r^2.$$

Thus $r/T_f(r) = 1$, so Theorem 5.8 does not apply.

Proof. The coordinate maximum is $\max(0, 2x)$, whose endpoint mean is r . The function $8x^2$ has only a second-order root at 0, of multiplicity 16; the factor $1/2$ in the integrated count gives $8r^2$. $\square$

Formalization map. [example59_ambientReciprocalCounting](#) in [Curves/FermatExamples.lean](#).

Example (5.10 – Sharp constants). Define

$$f_1(x) = \begin{cases} 0, & x \leq 1, \\ (2k-1)x - (2k^2-1), & 2k-1 \leq x < 2k+1, k \geq 1, \end{cases}$$

and

$$f_2(x) = \begin{cases} 0, & x \leq 2, \\ 2kx - 2k(k+1), & 2k \leq x < 2k+2, k \geq 1. \end{cases}$$

For $g = [0 : f_1]$, $h = [f_1 : f_2]$, and $P = x_0 + x_1$,

$$\begin{aligned} T_g(r) &= N^{(1)}(r, -P \circ g) = \frac{1}{4}r^2 + o(r^2), \\ T_h(r) &= \frac{1}{4}r^2 + o(r^2), \quad N^{(1)}(r, -P \circ h) = \frac{1}{2}r^2 + o(r^2). \end{aligned}$$

Thus the lower and upper constants in Theorem 5.8 are sharp.

Proof. Directly on the defining intervals, $f_1(r) = r^2/2 + O(r)$, $f_2(r) = r^2/2 + O(r)$, and their maximum has the same asymptotic; all three functions vanish on the negative tail. Hence the Cartan endpoint average for either curve is $r^2/4 + o(r^2)$. The functions f_1 and $f_1 + f_2$ are entire, so Jensen identifies their reciprocal counts with their endpoint means. These are respectively $r^2/4 + o(r^2)$ and $r^2/2 + o(r^2)$, proving both assertions. $\square$

Formalization map. [example510_concrete_conclusions](#) in [Curves/FermatExamples.lean](#).

6 Casoratian truncation

Definition (Shifts, partial counts, and Casoratian). For $k \geq 0$, put $\bar{g}^{[k]}(x) = g(x + k)$. For a Borel interval A , the partial integrated count is

$$N_A^{(j)}(r, f) = \frac{1}{2} \sum_{\substack{x \in A \cap (-r, r) \\ \omega_f^{(j)}(x) < 0}} |\omega_f^{(j)}(x)| (r - |x|)^j.$$

For entire $g_0, \dots, g_m$, define

$$C_0(g_0, \dots, g_m)(x) = \max_{\pi \in S_{m+1}} \sum_{i=0}^m g_i(x + \pi(i)).$$

Theorem (Finite-root first-order identity). *Let $f = [f_0 : \dots : f_m]$ be a first-order tropical holomorphic curve whose coordinates have finite root sets, possibly empty. Its Casoratian $C_0 = C_0(f_0, \dots, f_m)$ is entire of order at most one, has finitely many roots, and may be constant. Write the eventual slopes as $k_i^\pm, k_C^\pm$ and put $a = \frac{1}{2} \sum_i (k_i^+ - k_i^-) \geq 0$. Then*

$$k_C^+ - k_C^- = \sum_i (k_i^+ - k_i^-),$$

and there are constants M_i, M_C such that, for all sufficiently large r ,

$$N(r, -f_i) = \frac{k_i^+ - k_i^-}{2} r - M_i, \quad N(r, -C_0) = ar - M_C.$$

Consequently

$$N(r, -C_0) - \sum_{i=0}^m N(r, -f_i) = O(1),$$

and there is a function $\varepsilon(r) \rightarrow 0$ such that, eventually,

$$N(r, -C_0) = \left(\sum_{i=0}^m N(r, -f_i) \right) (1 + \varepsilon(r)).$$

Here the minus signs indicate root counts, using the paper's pole-counting convention for $N(r, h)$.

Proof. Choose a positive bound beyond all coordinate roots; this also makes sense when some or all root sets are empty. On either tail, the first-order multiplicity, hence every change of slope, vanishes. Continuity gives affine tails for each coordinate. Every permutation summand defining C_0 has right slope $\sum_i k_i^+$ and left slope $\sum_i k_i^-$; their finite maximum has the same tail slopes. Thus all Casoratian roots lie in a bounded interval, and local finiteness makes their set finite.

Jensen applied at two radii beyond all roots identifies total root multiplicity with the difference of the tail slopes. Put

$$M_i = \frac{1}{2} \sum_{x \in X_i} \omega_{f_i}(x) |x|, \quad M_C = \frac{1}{2} \sum_{x \in X_C} \omega_{C_0}(x) |x|,$$

where X_i, X_C are the finite root sets. Expanding the integrated counts gives the exact affine formulas. Their difference is the constant $\sum_i M_i - M_C$, proving the $O(1)$ estimate.

If $a > 0$, the sum of coordinate root counts grows linearly, so the constant difference divided by that sum tends to zero. If $a = 0$, nonnegativity of all root multiplicities forces every coordinate and Casoratian root count to vanish; take $\varepsilon = 0$. No division by a zero count is needed. $\square$

Formalization map. `FiniteFirstOrderRootData.ofFiniteRoots` constructs affine tails from finite root sets in `Truncated/FiniteRoots.lean`. The curve-facing statements `finiteRootCasoratian_countingDifference_isBigO_one`, `finiteRootCasoratian_firstOrder_identity`, and `finiteRootCasoratian_firstOrder_multiplicative` in `Truncated/ShiftCounting.lean` construct the Casoratian data internally.

Lemma (6.1 – Casoratian common factor). *For tropical entire $f_0, \dots, f_m, g$,*

$$C_0(f_0 \otimes g, \dots, f_m \otimes g) = g \otimes \bar{g} \otimes \dots \otimes \bar{g}^{[m]} \otimes C_0(f_0, \dots, f_m).$$

Proof. Expand the Casoratian as the maximum over permutations. Every permutation uses each shift $0, \dots, m$ exactly once, so the shifted copies of g factor out as the same additive term in every summand. Maximum commutes with adding this common term. $\square$

Formalization map. `tropicalCasoratian_commonFactor` and `casoratian_commonFactor_of_entire` in `Truncated/Casoratian.lean`.

Theorem (6.2 – Truncated second main theorem). *Let $f = [f_0 : \dots : f_m]$ be a nonconstant reduced n -th tropical holomorphic curve. Suppose that for some ℓ every $f_i - f_\ell$ is well-defined, that $\rho_2(T_f) = \rho_2 < 1$, and that $0 < \sigma < 1 - \rho_2$. Then, outside a set of finite logarithmic measure,*

$$\begin{aligned} & \left| \sum_{i=0}^m \sum_{j=1}^n N^{(j)}(r, -f_i) - \sum_{j=1}^n N^{(j)}(r, -C_0(f_0, \dots, f_m)) \right| \\ & \leq \sum_{j=1}^n \left| N^{(j)}(r, C_0(f_0, \dots, f_m)) - \sum_{i=0}^m N^{(j)}(r, \bar{f}_i^{[i]}) \right| \\ & \quad + \sum_{i=0}^m \sum_{j=1}^{\lfloor n/2 \rfloor} \left| N_{[-2i,0]}^{(2j)}(r, -\bar{f}_i^{[i]}) - N_{[-i,i]}^{(2j)}(r, -f_i) \right| + o\left(\frac{T_f(r)}{r^\sigma}\right). \end{aligned}$$

Proof. For a root x of f_i , translation sends it to $x - k$. The signed multiplicity is unchanged for odd j and whenever x and $x - k$ have the same sign. Thus only even orders crossing the origin contribute a defect, which is exactly

$$N_{[-2k,0]}^{(2j)}(r, -\bar{f}_i^{[k]}) - N_{[-k,k]}^{(2j)}(r, -f_i).$$

Partition the remaining roots into the central region and the four boundary strips determined by $r - k, r, r + k$. Monotonicity of root counting sandwiches their total contribution between the corresponding counts at $r - k$ and $r + k$. Proposition 5.4 dominates every coordinate quotient characteristic by $T_f + O(1)$; Lemma 4.6 then makes each fixed-radius increment $o(T_f/r^\sigma)$. Summing the finitely many i, j, k proves the shifted-count identity used in the displayed right side.

By Lemma 6.1, subtracting the reference coordinate f_ℓ from all coordinates only adds the common shifted factor $\sum_{k=0}^m f_\ell(x + k)$ to the Casoratian. At either endpoint δr , choose a maximizing permutation π_δ . Theorem 4.7, applied to each well-defined quotient $f_i - f_\ell$, gives

$$\left| C_0(f_0, \dots, f_m)(\delta r) - \sum_{i=0}^m \bar{f}_i^{[i]}(\delta r) \right| = o(T_f(r)/r^\sigma).$$

Indeed, after removing the common factor the difference is bounded by the finite sum of shift differences between $\pi_\delta(i)$ and i .

Finally apply the n -th Jensen identity to C_0 and separately to every $\bar{f}_i^{[i]}$. Insert the shifted-count identity from the first paragraph and the endpoint estimate from the second, then apply the triangle

inequality. This is exactly the asserted inequality. Nonconstancy, convexity, and Proposition 5.4 imply the characteristic scale tends to infinity and absorb the finitely many $O(1)$ terms; no additional growth assumption is used. $\square$

Formalization map. `truncatedSecondMain` and `truncatedSecondMainPaperStatement_verified` in `Truncated/Main.lean`; shift transport and the exact even correction are in `Truncated/ShiftCounting.lean`.

Example (Higher-order correction terms are necessary). Let

$$f_0(x) = \begin{cases} -x^2, & x \leq 0, \\ x^2, & x > 0, \end{cases} \quad f_1(x) = \begin{cases} x^2, & x \leq 0, \\ -x^2, & x > 0. \end{cases}$$

Then $f = [f_0 : f_1]$ is a well-defined second-order curve and

$$C_0(f_0, f_1)(x) = \begin{cases} -2x - 1, & x \leq -1, \\ 2x^2 + 2x + 1, & -1 < x \leq 0, \\ 2x + 1, & x > 0. \end{cases}$$

Moreover $T_f(r) = r^2$, the coordinate reciprocal counts vanish, while

$$\sum_{j=1}^2 N^{(j)}(r, -C_0) = r^2.$$

Hence the two explicit error sums in Theorem 6.2 cannot in general be replaced by $o(T_f/r^\sigma)$ when $n \geq 2$.

Proof. The coordinate maximum is x^2 , giving $T_f(r) = r^2$. The negatives of both coordinates are second-order entire, so their reciprocal counts are zero. Taking the maximum of the two Casoratian permutation sums yields the displayed three pieces. Its only reciprocal-count contribution is the normalized second-order root at 0; direct substitution gives integrated count r^2 . $\square$

Formalization map. `section6Counterexample_exact_obstruction` in `Counterexamples.lean`.

Corollary (6.3 — First-order truncated relation). *For a nonconstant reduced first-order curve satisfying the same hyper-order condition,*

$$\sum_{i=0}^m N(r, -f_i) = N(r, -C_0(f_0, \dots, f_m)) + o\left(\frac{T_f(r)}{r^\sigma}\right)$$

outside a set of finite logarithmic measure.

Proof. At first order there is no even-order correction. The shifted coordinate functions and the Casoratian are tropical entire, so the two explicit pole count errors in Theorem 6.2 vanish. The remaining term is the stated little- o remainder. $\square$

Formalization map. `firstOrderTruncatedSecondMainPaperStatement_verified` in `Truncated/Main.lean`.

Example (The hyper-order condition in Corollary 6.3 is necessary). Let $f = [0 : e_\alpha]$ with $\alpha > 1$. Then $T_f(r) = T(r, e_\alpha) + O(1)$ has hyper-order 1, $e_\alpha(x+1) = \alpha e_\alpha(x)$, and

$$C_0(0, e_\alpha) = \alpha e_\alpha.$$

Consequently

$$\begin{aligned} N(r, -0) + N(r, -e_\alpha) &\sim T(r, e_\alpha), \\ N(r, -C_0(0, e_\alpha)) &\sim \alpha T(r, e_\alpha), \end{aligned}$$

so the conclusion of Corollary 6.3 fails.

Proof. The shift equation follows directly from the floor-function definition of e_α . Of the two Casoratian permutation sums, nonnegativity makes the shifted hyper-exponential the maximum, hence the Casoratian equals αe_α . Proposition 3.8 gives hyper-order 1. Jensen gives the first asymptotic, and positive scalar homogeneity of multiplicities multiplies the second count by α . Since $\alpha > 1$, the two leading terms cannot differ by the little- o remainder required in Corollary 6.3. $\square$

Formalization map. `corollary63_CasoratianReciprocalCounting_isEquivalent_scaledCharacteristic` and `corollary63_growthHypothesis_counterexample` in `Counterexamples.lean`.